



PROJECT

TPR2251-PATTERN RECOGNITION

TRIMESTER March 2024, 2023/2024 (Term 2410)

Project Title: Implementation of Parts Recognition and Counting System

Lab Section: 1C

Submission Date: 24 June 2024

NO	STUDENT ID	STUDENT NAME	MAJOR
1.	1211105012	YEO QUAN XIANG	AI
2.	1221301165	JASON TAN KHIM CHEN	AI
3.	1211309854	ABDULLA MOHAMED ALHEJAIRI	AI

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Introduction

Our group task is developing a model that is helpful for part recognition and stock counting for TSH Group Pte Ltd. Currently, the TSH Group carries out stock counting and part recognition manually, and there might be a higher possibility of error if compared to machine operation to count the stock.

Consequently, by implementing automation for stock counting and part recognition, our program aims to increase the accuracy, efficiency, and effectiveness of tasks like stock counting and part recognition for TSH Group.

To implement the automation in stock counting and part recognition for TSH Group, our group has trained two models. The first model is trained in part recognition. It preprocesses the images captured using preprocessing steps such as gray scaling, segmentation, resizing, normalization and flattening. Feature extraction used in the model is the Linear Discriminant Analysis (LDA), and the machine learning algorithm used is K-Nearest Neighbors (KNN). We will also implement a grid search to find the best parameter to be tuned for the model, achieving the best accuracy. After evaluating the model, our group will visualize the results in the confusion matrix and ROC curve for better data interpretation and analysis by our group members and better model performance understanding by the TSH Group.

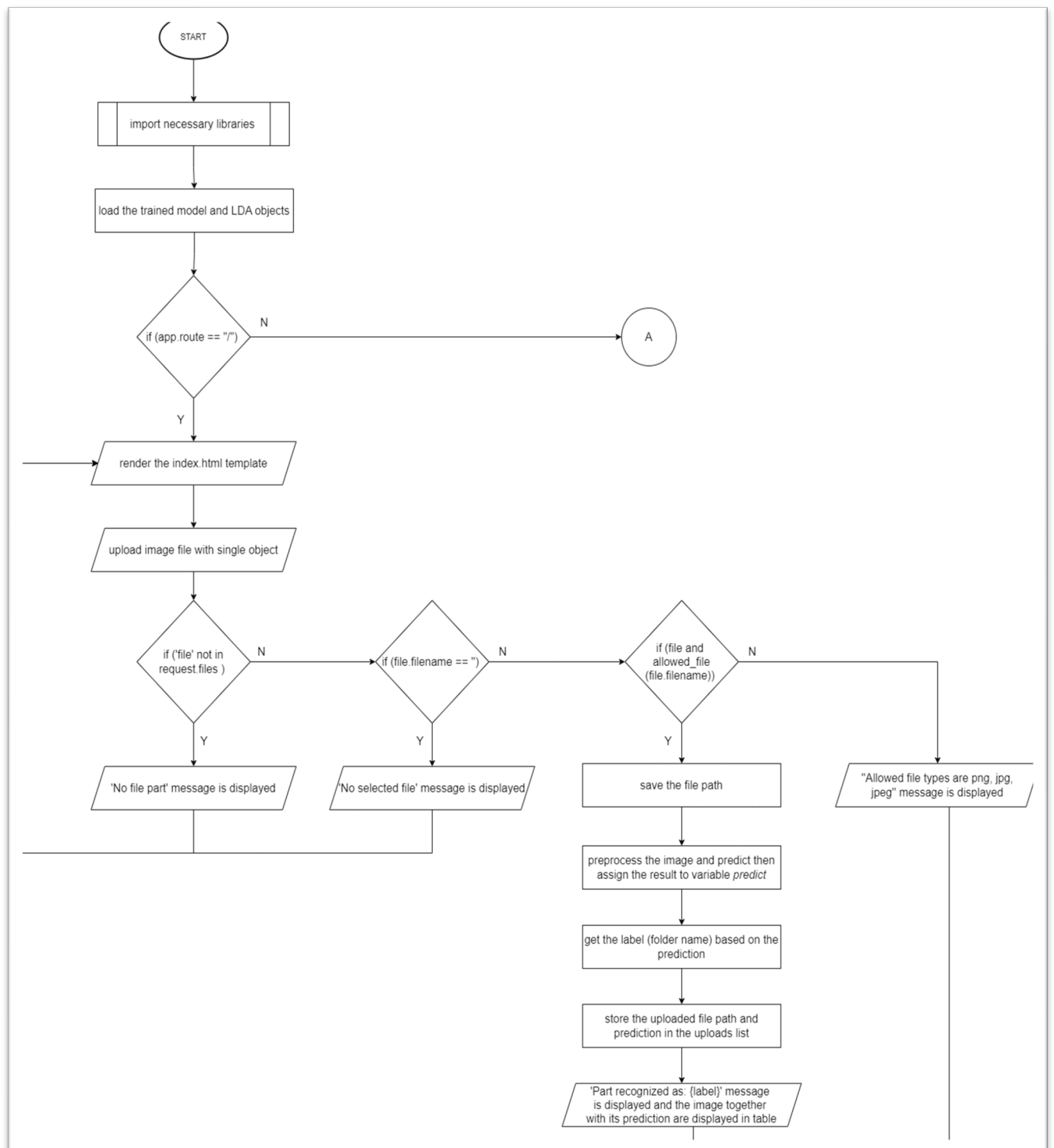
The second model, Roboflow model is trained in counting algorithms. First, it annotates the images uploaded and moves them to datasets. The datasets will be split into training set, validation set, and test set in a ratio of 8:1:1. The images are preprocessed using steps like resizing, gray scaling and auto orientation. Augmentation is also applied to the images, such as flip: horizontal and rotation: 15 degrees plus minus. The model will then train with the datasets and the performance of the model will be evaluated in mAP, precision and recall, and visualized in graphs.

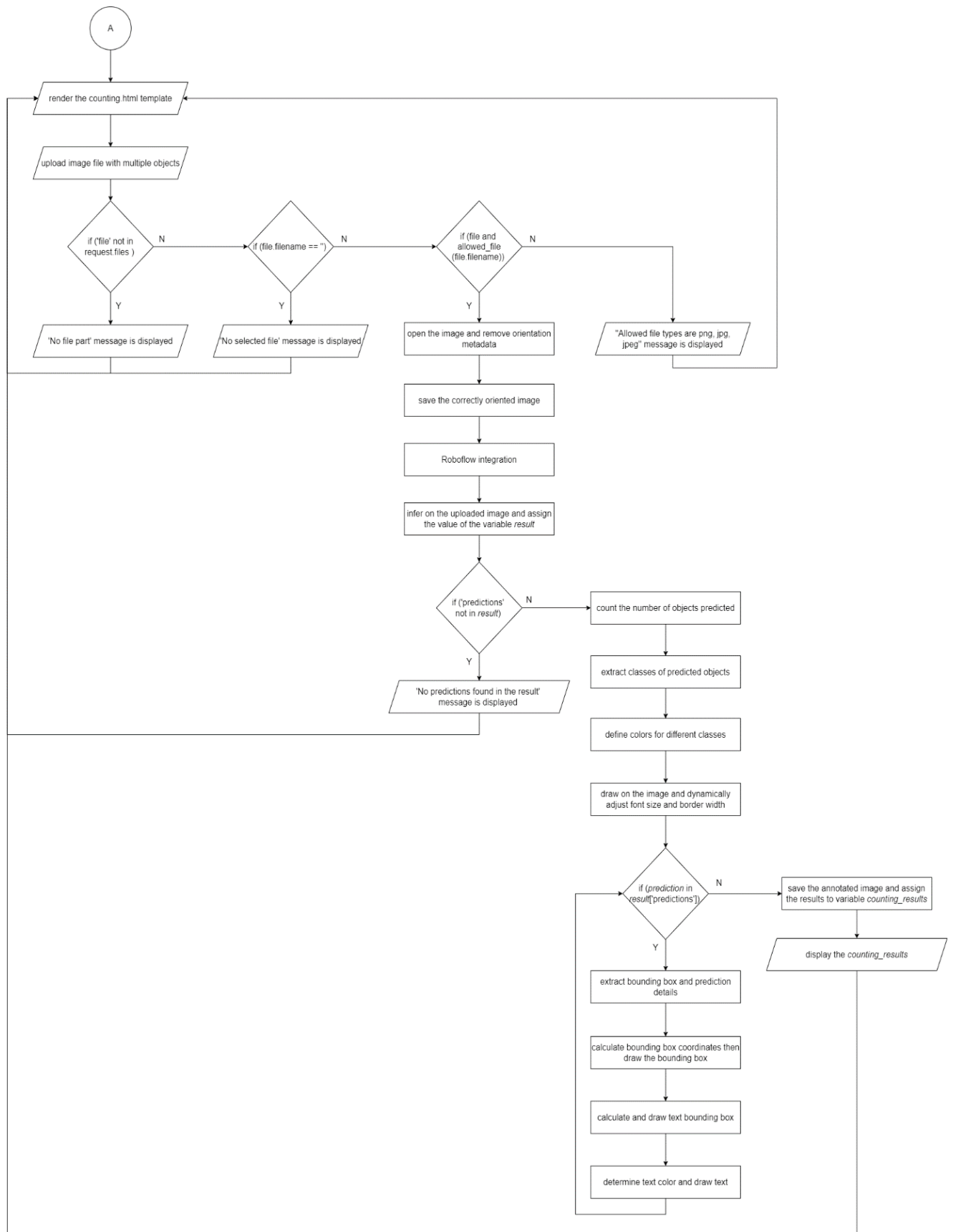
In a nutshell, both models will be implemented in the graphical user interface (GUI) that works as a program to help the TSH Group to recognize parts and count the stock automatically instead of manual operation by uploading the images to the GUI. The results will be displayed in table form, which provides better visualization.

Design and Implementation

app.py

Flow chart of system





First, the system will import all the necessary libraries and load the trained model and LDA objects. The user will then be directed to the index.html by default; if the user clicks on the counting navigation link, it will direct to the counting.html.

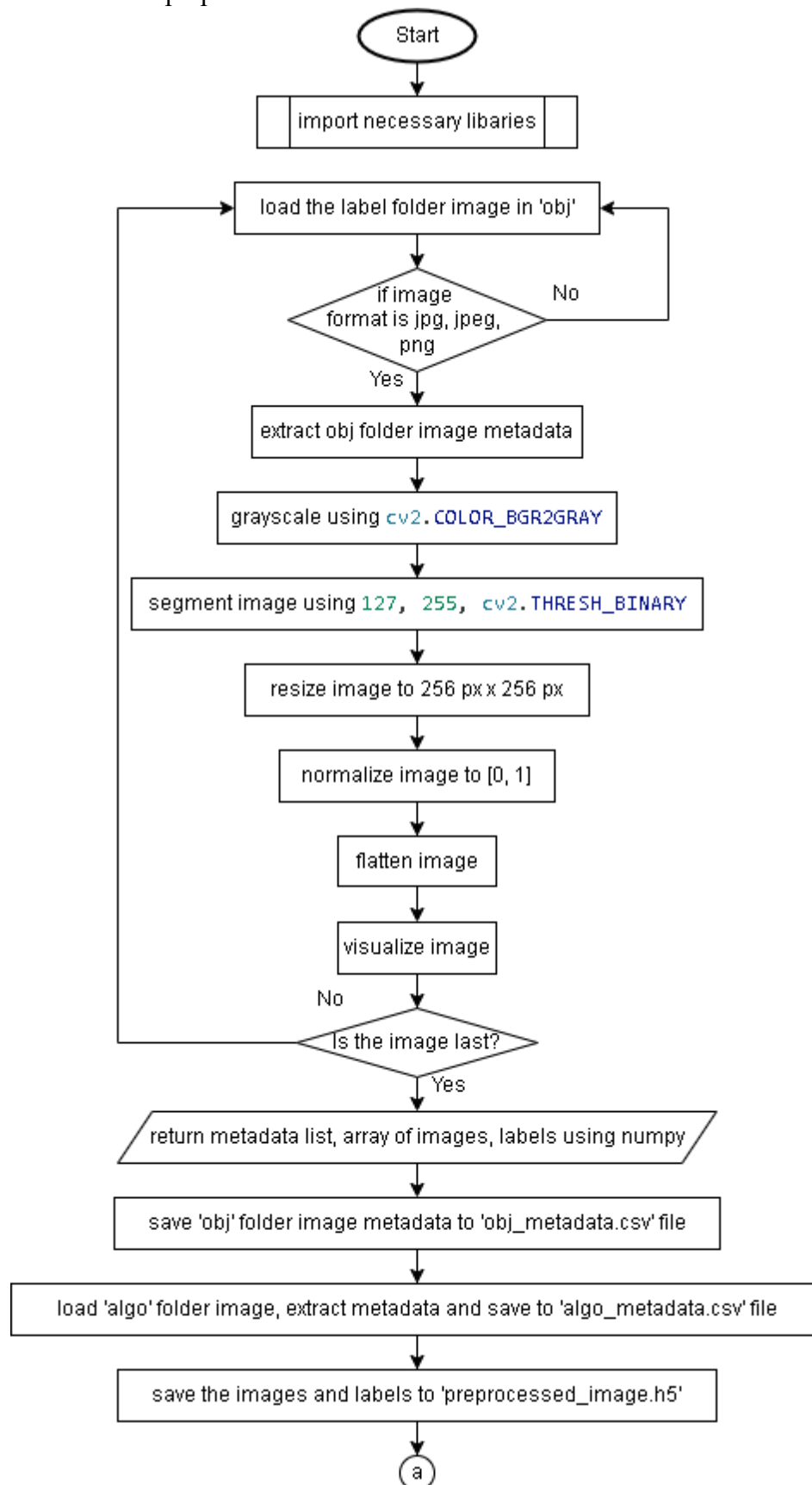
In index.html, the program recognizes objects based on images uploaded by users. They can upload an image with a single object. The uploaded image will be preprocessed and predicted if it meets the requirements. The results will be displayed in table form.

In counting.html, the program counts the objects available in the image uploaded by the user. They can upload an image with multiple objects, and if the uploaded image meets the requirements, it is open, and its orientation metadata will be removed. After orienting the image correctly, the Roboflow model is integrated, and this model is a unique feature for the counting algorithm because it provides a user interface for annotating, preprocessing, and augmenting the images. The results of inferring the uploaded images will be returned by the Roboflow model in JSON format with the name "predictions" and corresponding values.

If the "predictions" name is not detected in the result, meaning no prediction was made to the uploaded image and the 'No predictions found in the result' message will be displayed; else, the model will count the number of objects predicted, extract classes of predicted objects, define colors for different classes and apply on the uploaded images. Then, looping for the prediction will be carried out by extracting the bounding box and prediction details, calculating the bounding box coordinates, then drawing the bounding box, followed by calculating and drawing the text bounding box until all the values in predictions are being looped. The annotated image, object count, and object classes will be displayed in table form at the end.

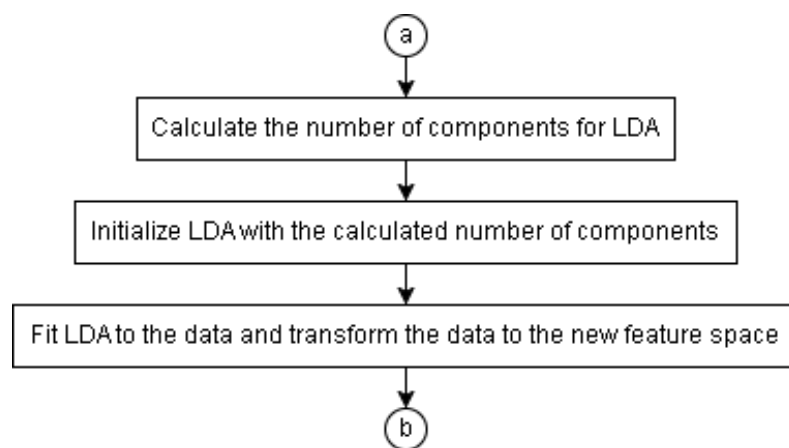
model.py

Flow chart of preprocess



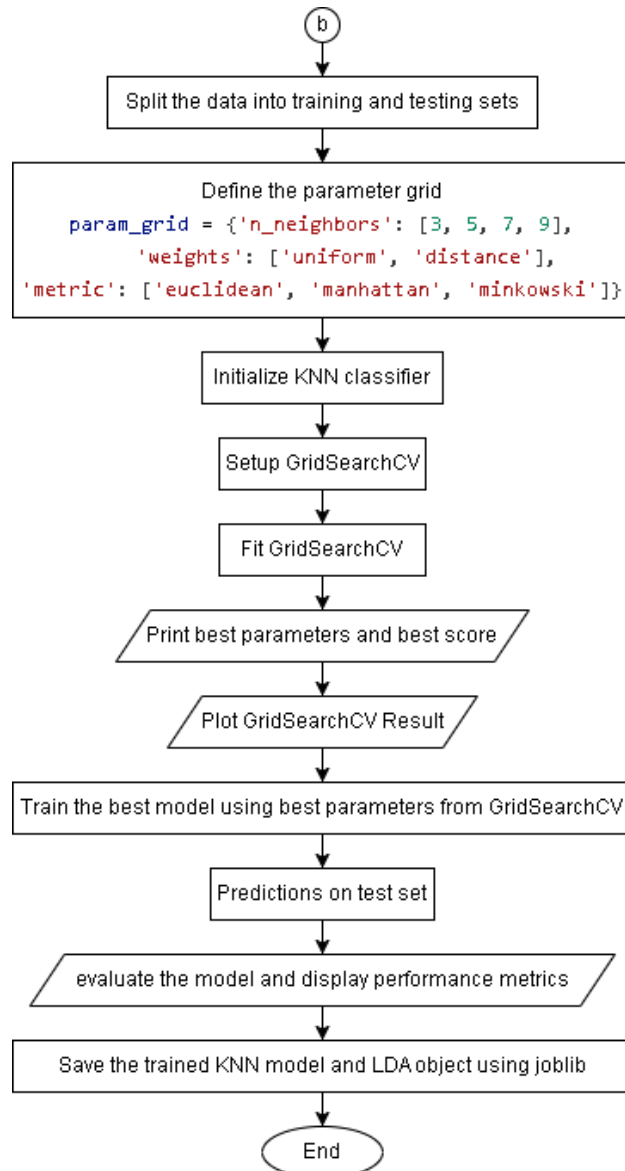
First, the system will import all the necessary libraries, followed by the execution of looping by loading the label folder image in the obj folder. The system will then check if the image format meets the requirements: if yes, the obj folder image metadata will be extracted, followed by gray scaling, segmenting, resizing, normalizing, flattening, and visualizing them; else, the label folder image will be loaded again. The loop will continue until the last image, and all the metadata list, an array of images, and labels will be returned, and the metadata will be saved to the 'obj_metadata.csv' file. Then, the algo folder image will be loaded to extract the images' metadata inside and saved to the algo_metadata.csv file. The preprocess steps end with saving the images and corresponding labels to the 'preprocessed_image.h5' file.

Flow chart of feature extraction



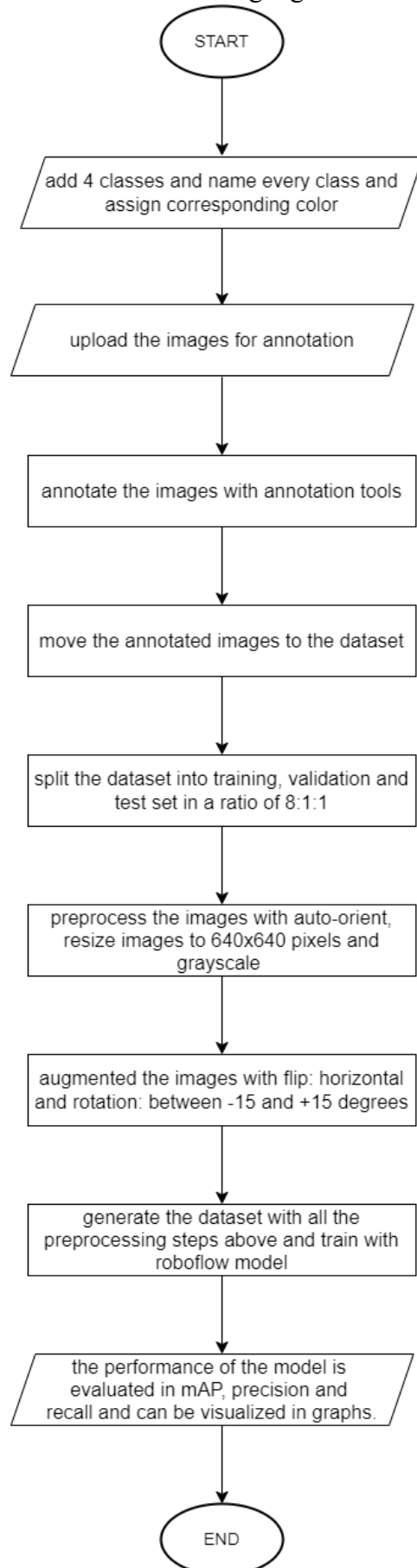
After preprocessing, we carry out feature extraction on the images and corresponding labels using LDA. First, we calculate the number of components for LDA, then initialize LDA with them, fit LDA to the data, and transform the data to the new reduced feature space.

Flow chart of classification



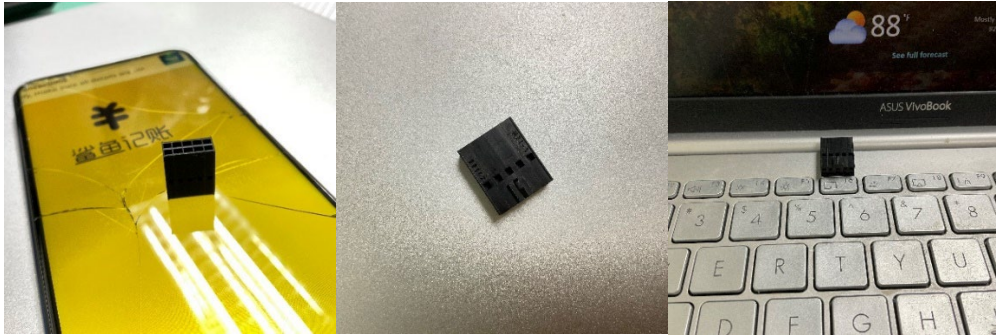
After feature extraction, the data is split into training and testing sets. The parameter grid is defined for grid search purposes later. We will initialize the KNN classifier, set up GridSearchCV, and fit it into the parameter grid. The best parameters and score will be printed, and the result will be plotted. The best model is then trained using the best parameters from the grid search; the model predictions are then made on the test set. Lastly, the model will be evaluated, and the performance metrics displayed. The trained KNN model and LDA object are exported using joblib, and the model training and testing are complete.

Flow chart of counting algorithm

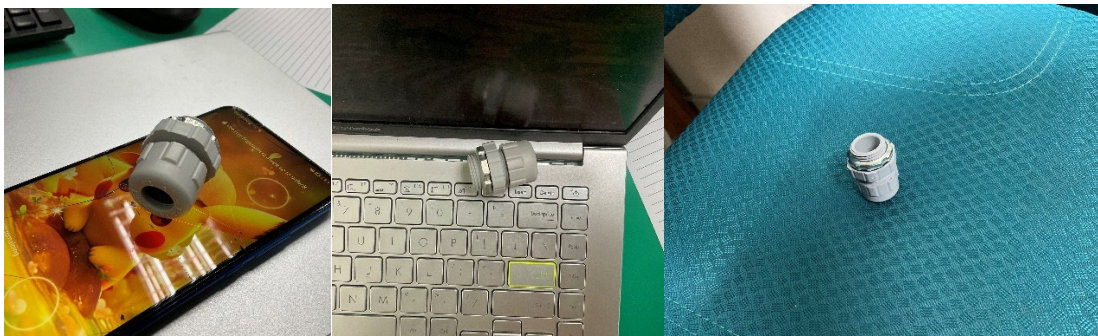


For the algorithm used in counting the number of objects that appeared in the images, our group has implemented the Roboflow model, and the following are the steps in the Roboflow model. First, we add four classes and assign the names and colors for every class. Then, we will upload the images for annotation. We annotate the images using manual labeling but with assistance from AI labeling. After annotating the images, we move them to the dataset and split them into training, validation and test sets. We then choose the preprocessing methods and augmentation for our images, generating the final dataset to be trained by the model. The model is then used to train the preprocessed dataset, and the performance of the model will be evaluated and visualized.

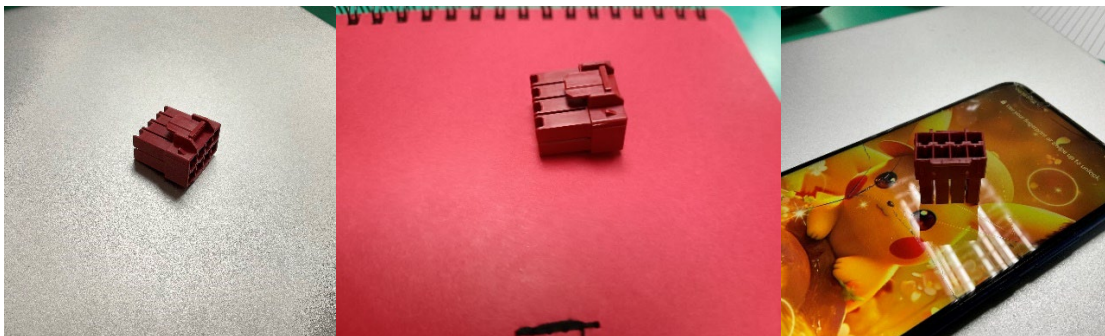
Description of input data



Black box plastic connector, Rectangle prism with a series of small squares cut out of it.



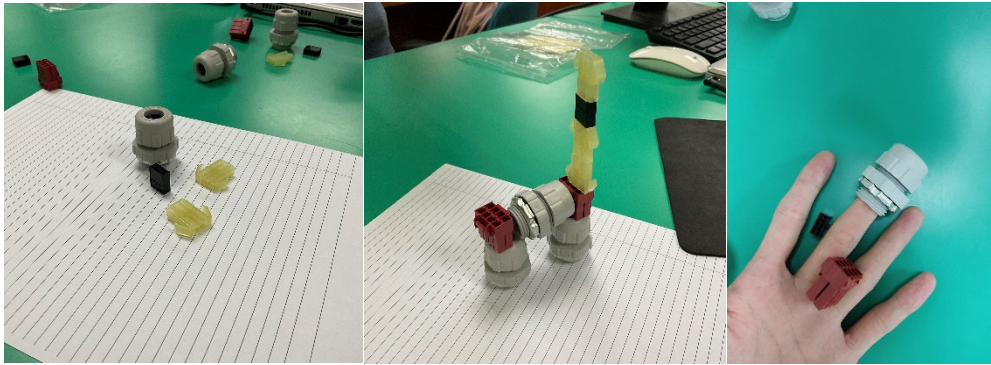
Grey hollow cylindrical plastic connector with metal ring on top.



Red rectangular plastic connector. Two prongs at one end.



Yellow connector. With 3 holes and 2 ears.



Counting algorithm image with multiple arrangement and placement.

Link to images folder:

[https://mmuedumy-my.sharepoint.com/:f:/r/personal/1211105012_student_mmu_edu_my/Documents/Tpr project/image?csf=1&web=1&e=qglAOi](https://mmuedumy-my.sharepoint.com/:f:/r/personal/1211105012_student_mmu_edu_my/Documents/Tpr%20project/image?csf=1&web=1&e=qglAOi)

Experiment Setup

The number of images collected for the two models is 916, whereas 480 images are for the model for part recognition and 436 images for the Roboflow model for the counting algorithm. The images were collected using Redmi Note 11 Pro+ 5G, POCO X3 Pro, and Apple iPad Air (5th generation). The camera specifications of the images collected for both models are as follows:

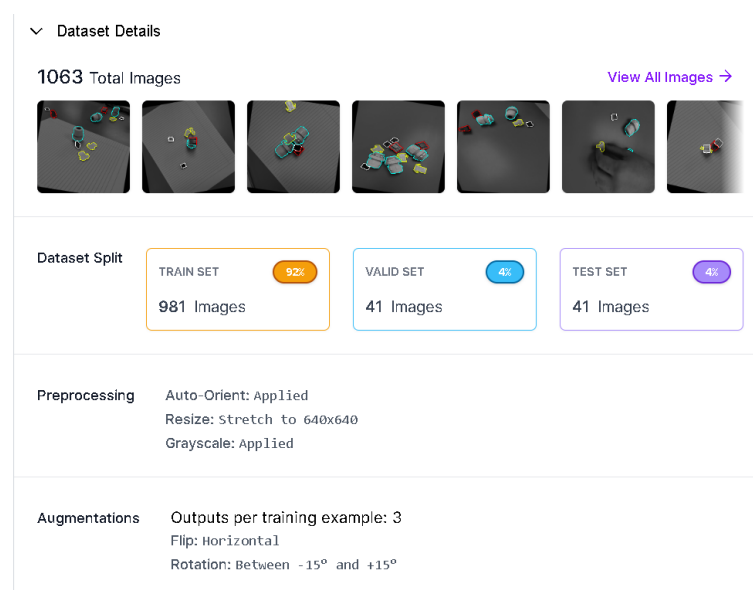
- **Make:** Xiaomi and Apple.
- **Model:** Redmi Note 11 Pro+ 5G, POCO X3 Pro, Apple iPad Air (5th generation).
- **Orientation:** Normal orientation, rotated 180 degrees, rotated 90 degrees clockwise, rotated 90 degrees counter-clockwise, mirrored horizontally, then rotated 90 degrees counter-clockwise and some images orientation information is either missing or not set correctly.
- **Metering mode:** Average, center-weighted average, spot
- **Lightsources:** D65 and other light source
- **Flash:** Flash did not fire, compulsory flash mode
- **Focal length:** ranged from 1.65mm to 5.89mm
- **Focal length in 35mm film:** 0mm, 25mm
- **Exposure bias value:** 0 EV
- **Shutter speed value:** ranged from 4.321s^{-1} to 6.906s^{-1}
- **Exposure time:** ranged from 0.008333s to 0.058337s
- **F-Number:** f/2.2, f/1.89, f/1.79
- **Exposure program:** Normal program, not defined
- **ISO Speed ratings:** ranged from 88 to 6333
- **Exposure mode:** Manual
- **Sensitivity type:** Unknown
- **White balance:** Auto and manual white balance

- **Sensing method:** Not defined

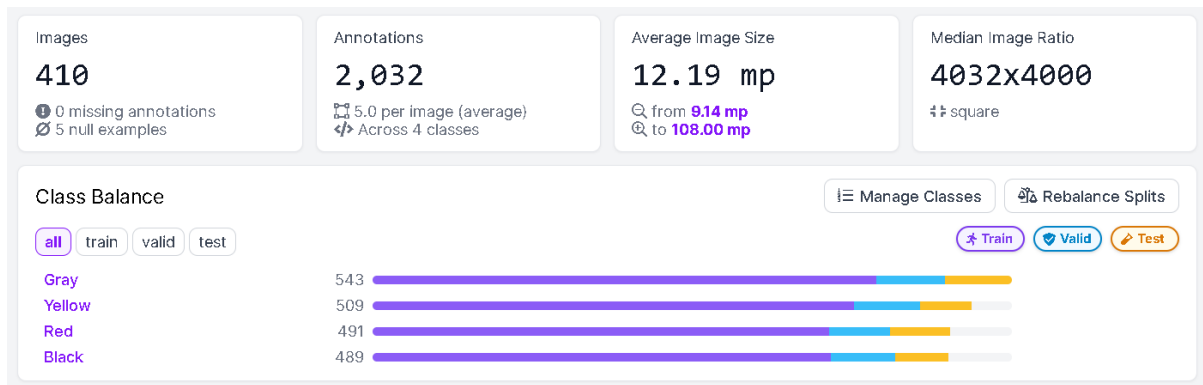
The image resolution of the images collected for both models is as follows:

- **Resolution unit:** Inches
- **Image width:** ranged from 2448 pixels to 12000 pixels
- **Image length:** ranged from 2448 pixels to 12000 pixels
- **XResolution:** 72 dpi
- **YResolution:** 72 dpi
- **Exif image width:** ranged from 2448 pixels to 12000 pixels
- **Exif image height:** ranged from 2448 pixels to 12000 pixels

The number of training images for the model used for parts recognition is 288, and the other 192 images are used to test, meaning the ratio of training to testing set is 6:4.



The number of training images for the Roboflow model used for the counting algorithm is 981, and the other 82 images are used to test. The training images for the Roboflow model increase because augmentation is applied to the training images, so some training images are duplicated with different augmentations, such as flip, horizontal, and rotation, 15 degrees plus-minus. However, we initially set the ratio of the training set, validation set, and test set to 8:1:1 based on the original number of images we collected, which is 436 images.



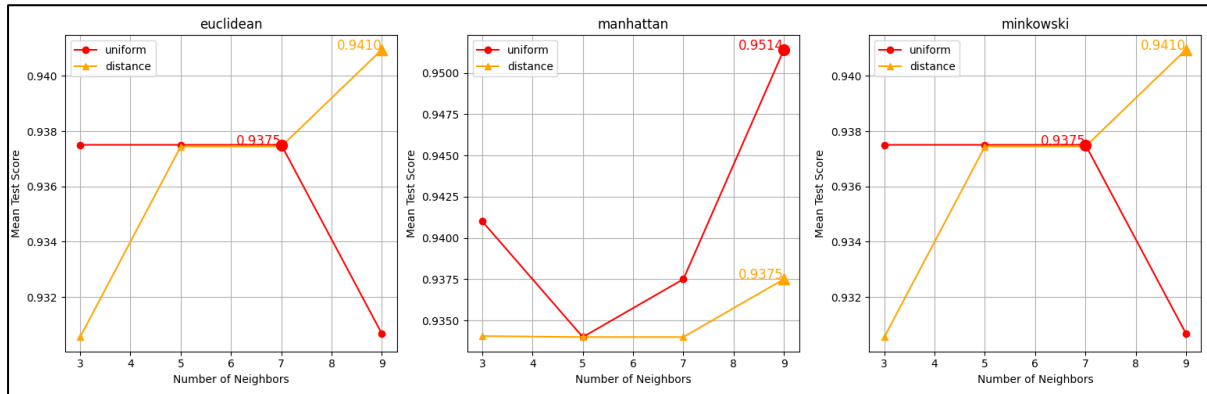
Performance Evaluation

Part Recognition Model: KNN

Fitting 5 folds for each of 24 candidates, totalling 120 fits

Best parameters: {'metric': 'manhattan', 'n_neighbors': 9, 'weights': 'uniform'}

Best cross-validation accuracy: 0.9514216575922566



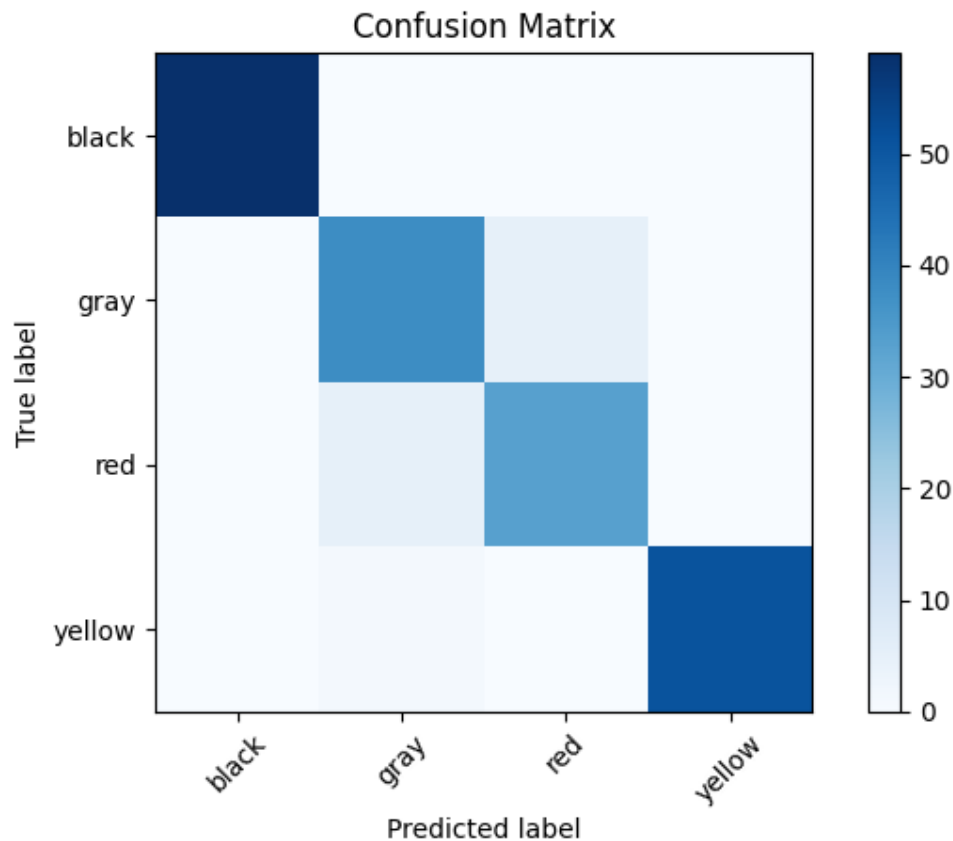
The figure above shows that the best parameters for KNN searched by the GridSearchCV for metric, n_neighbors and weights are Manhattan, 9 and uniform respectively.

Classification Report:				
	precision	recall	f1-score	support
black	1.00	1.00	1.00	59
gray	0.86	0.88	0.87	43
red	0.87	0.87	0.87	38
yellow	1.00	0.98	0.99	52
accuracy			0.94	192
macro avg	0.93	0.93	0.93	192
weighted avg	0.94	0.94	0.94	192
Accuracy Score: 0.9427083333333334				
Precision: 0.94				
Recall: 0.94				
F1-score: 0.94				

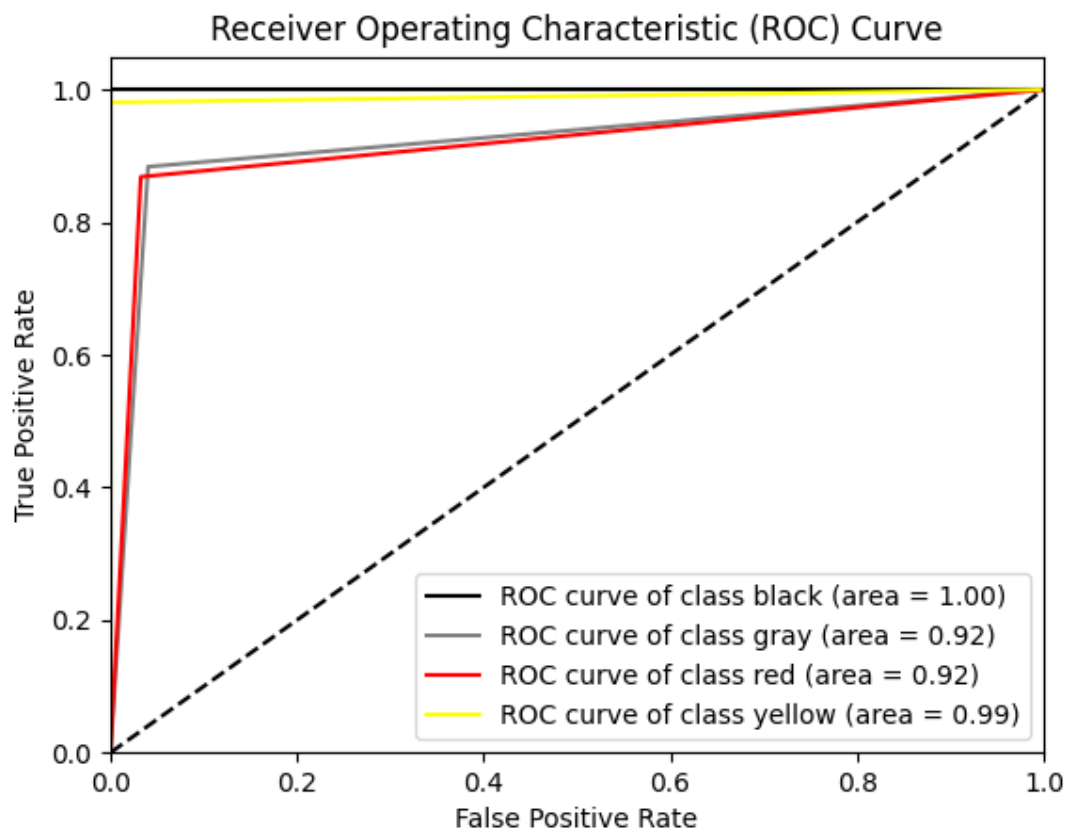
From the classification report of KNN above, the accuracy, precision, recall, and F1-score values are relatively high, at approximately 0.94. Among the four classes, the model can ideally recognize the class black, followed by the yellow, gray, and red.

Confusion Matrix:

```
[[59  0  0  0]
 [ 0 38  5  0]
 [ 0  5 33  0]
 [ 0  1  0 51]]
```



The KNN Confusion matrix visualizes the actual label and predicted label in matrix form and indicates the model's performance on the four classes with the intensity of color: The darker the box's color, the more predictions there are, and vice versa. From the confusion matrix above, we can conclude that class black has the most true predicted label, followed by class yellow, gray, and red.



From the KNN ROC curve above, class black has the highest Area Under the Curve (AUC), followed by class yellow, grey, and red. The greater the AUC, the better the model can make predictions. Hence, we can conclude that the model is the best in recognizing the class black, followed by class yellow, class gray, and class red.

Counting Algorithm Model: Roboflow Model**tpr-counting-algo/5**

Model Type: Roboflow 3.0 Object Detection (Fast)

Checkpoint: tpr-counting-algo/1

[More Metrics](#)[Visualize](#)mAP 
96.1%Precision 
93.8%Recall 
89.9%

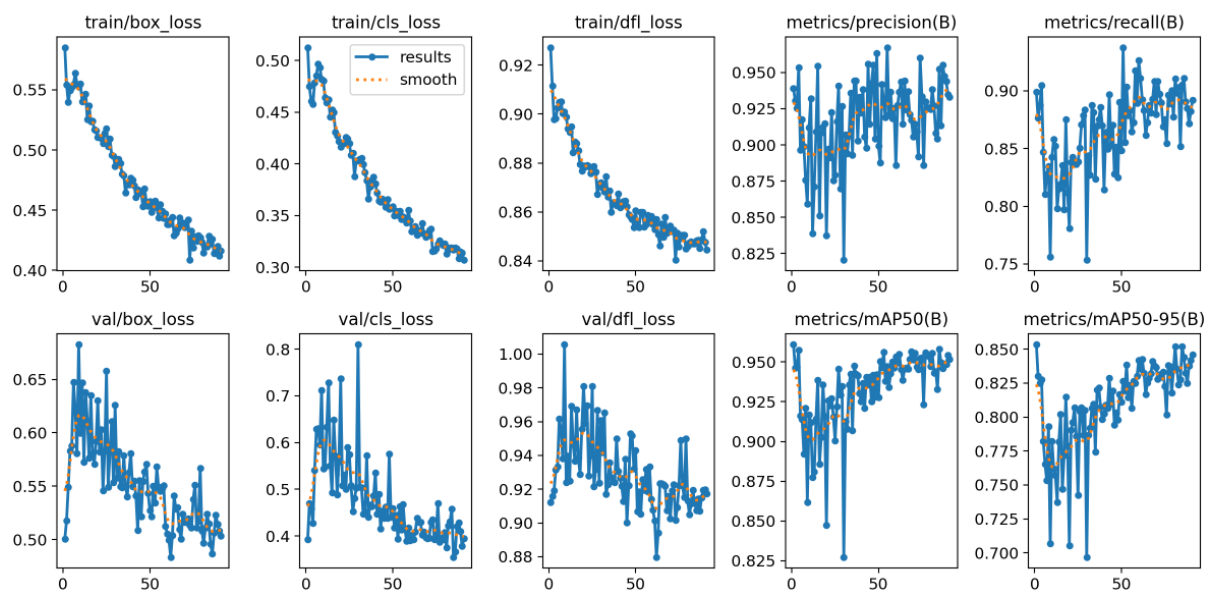
Validation Set Test Set Training Graphs

Average Precision by Class

Validation Set Test Set Training Graphs

Average Precision by Class

Results of fine-tuning training:





From the graphs above, we can interpret that the mean average precision (mAP) of the Roboflow model starts to stabilize and improves steadily after epoch 40. Box loss stands for mean square error, class loss stands for cross-entropy, and object loss stands for confidence in object presence. We can interpret that the 3 loss graphs continue to decline steadily as the epochs increase, meaning the model's performance becomes better and better.

Discussion and Conclusions

Discussion

In this project, we developed and implemented two models for TSH Group Pte Ltd to automate part recognition and stock counting. The program aims to improve the accuracy, efficiency, and effectiveness of these tasks, which are currently performed manually, by integrating both models into the program.

1. Model Execution:

Part Recognition Model:

We obtained good accuracy by combining many image preprocessing steps like grayscale, segmentation, scaling, normalization, and flattening. Linear Discriminant Analysis (LDA) was used for feature extraction, and K-nearest neighbours (KNN) for classification. A confusion matrix and ROC curves were used to evaluate the model, and it performed well in all classes—especially for the black-coloured portions.

Roboflow model:

Preprocessing and augmentation techniques were used on the model during the training, testing, and annotating phases to improve its performance. The model's accuracy in counting items was demonstrated by its steady improvement in precision, recall, and mean average precision (mAP) throughout epochs.

2. Technical Challenges:

Data Variability:

There was variation in the resolution, orientation, and device specs of the images uploaded for testing and training. This unpredictability made it challenging to guarantee reliable model predictions and uniform preprocessing.

Model Tuning:

Grid searches and evaluations were necessary to determine the ideal parameters for the part recognition and counting models, highlighting the significance of parameter tuning in attaining high performance.

3. Practical Implications:

The automated approach reduces the possibility of human error in stock counting and part recognition. The project's graphical user interface (GUI) ensures that users upload images and get results in an understandable, tabular style.

4. Future Improvements:

Improved Data Collection:

Raising training images' variety, quantity, and quality may improve the model's performance.

Real-time Processing:

Adding real-time processing capabilities could improve the system's effectiveness even more.

Integration with Inventory Systems:

The company's inventory management system could be connected to the recognition and counting system to enable smooth updates and real-time stock monitoring.

Conclusion

Automation of stock counting and parts recognition tasks has brought many benefits, as seen by the development and implementation of the parts recognition and counting system for TSH Group. In their respective tasks, the two models—one for part recognition and the other for counting—exhibited relatively high accuracy. In addition to achieving its goals, the research offered insightful information on how machine learning is applied in real-world industrial settings. Going forward, the system's performance and usefulness will be further enhanced by improvements in data diversity, real-time processing, and system integration.

We have obtained much professional and real-world experience in creating machine learning models through this project, including everything from feature extraction and data preprocessing to model training, assessment, and deployment. This research demonstrates how artificial intelligence (AI) can revolutionize conventional procedures and provide the groundwork for upcoming advancements in automated stock management.

References

State references you refer to complete your program

1. Roboflow. (2022, July 6). Getting Started with Roboflow (July 2022) [Video]. YouTube. <https://www.youtube.com/watch?v=O-ZPxTpb2Yg>
2. Flowchart Tutorial (with Symbols, Guide and Examples). (n.d.). <https://www.visual-paradigm.com/tutorials/flowchart-tutorial/>
3. Bobbitt, Z. (2021, August 9). How to interpret a ROC curve (With Examples). Statology. <https://www.statology.org/interpret-roc-curve/>
4. Arie, L. G., PhD. (2023, February 16). The practical guide for Object Detection with YOLOv5 algorithm. Medium. <https://towardsdatascience.com/the-practical-guide-for-object-detection-with-yolov5-algorithm-74c04aac4843>

Task Declaration Form																	
<b style="color: red;">Implementation of Parts Recognition and Counting System by Group 5 (YEO QUAN XIANG)																	
Group Member's Name	:	YEO QUAN XIANG															
Group Member's ID	:	1211105012															
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